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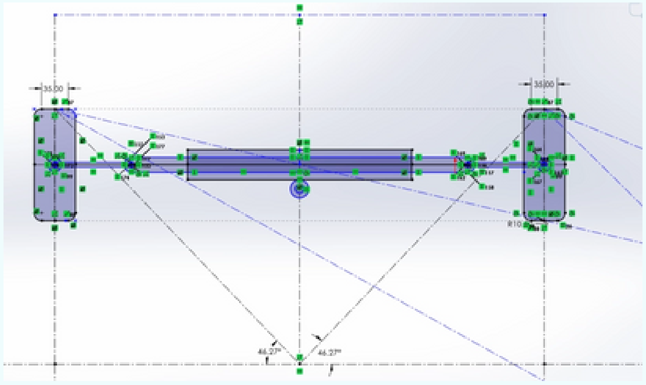
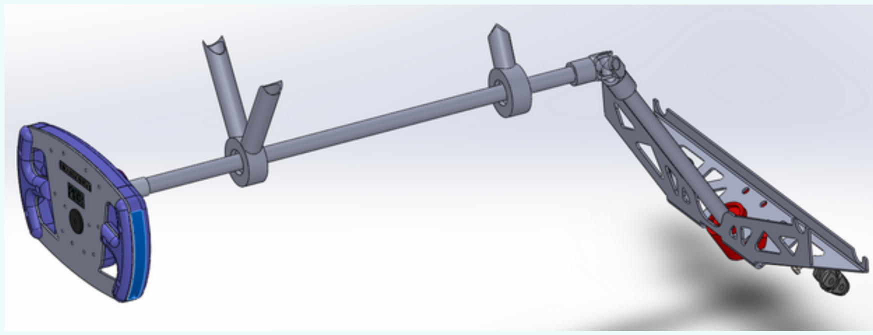
FSAE Rack and Pinion

Rack and Pinion Design





What?

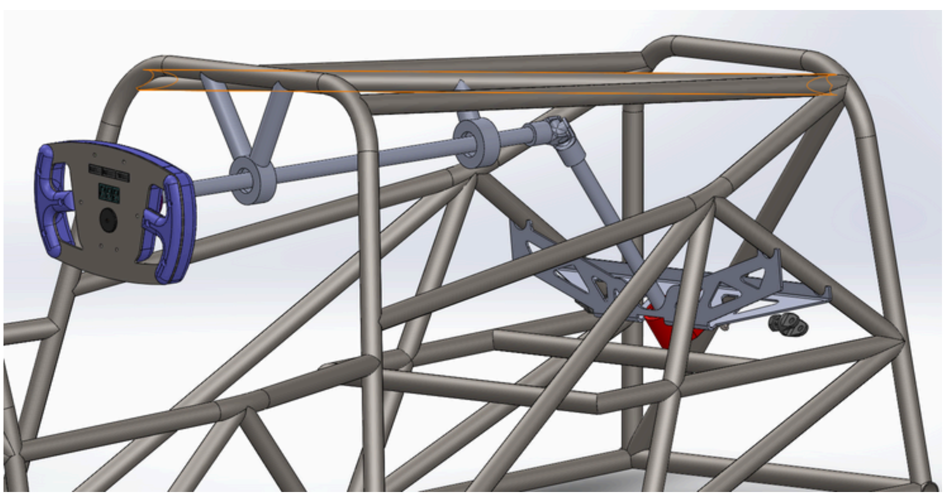
- Designed a **rack and pinion control system** integrated with a steering angle sensor to better the steering accuracy and vehicle dynamics in the Formula SAE vehicle.
- Developed a **closed-loop control system**, where the **driver's input** and **sensor feedback** continuously adjust the steering for **real-time corrections.**
- The project focused on achieving **precise control** through **feedback mechanisms** using the **Raspberry Pi** as a central control unit.

How?

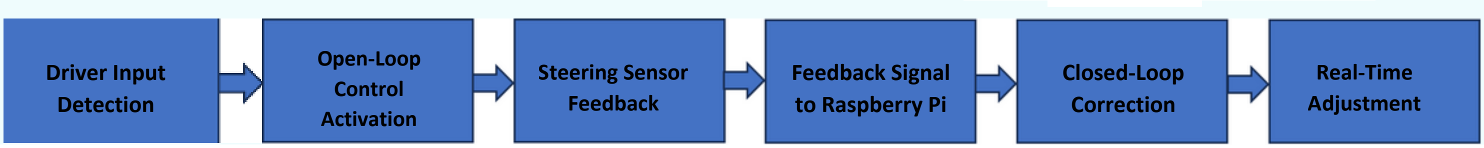
- Modeled the rack and pinion system using SolidWorks, ensuring proper integration with the steering sensor for **feedback control.**
- Implemented an **open-loop system** for initial steering control, where the **driver's input directly affects the steering angle without feedback.**
- **Transitioned** to a **closed-loop control system** by **integrating** a **Raspberry Pi** to process data from the steering sensor. The **feedback** from the sensor allows **real-time corrections** to the steering angle based on the driver's input
- **Manufactured components with high-precision** techniques to ensure durability, enabling the system to handle feedback-based adjustments under high loads.

Results

- The **closed-loop control system** significantly improved **steering precision** and **vehicle stability** by **continuously adjusting** the steering angle based on **feedback** from the steering sensor and **driver inputs.**
- The Raspberry Pi effectively processed **real-time sensor data**, reducing steering errors and enhancing overall handling.
- The system provided a smooth **transition** from **open-loop to closed-loop control**, demonstrating better performance, reliability, and responsiveness during testing.



Rack and Pinion System Control Manufacturing Process Flow



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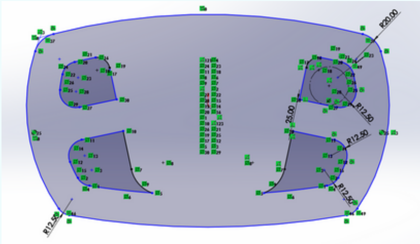
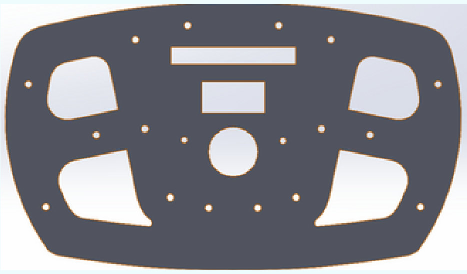
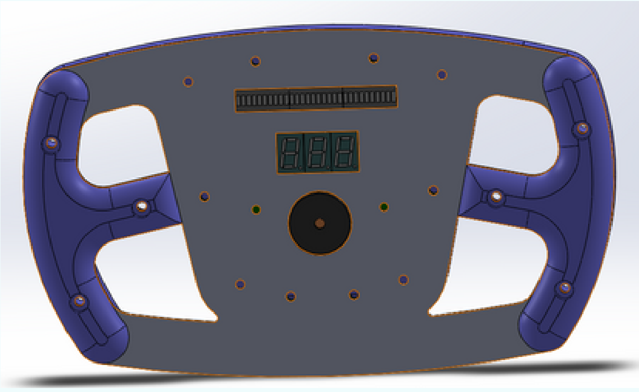
Formula SAE Steering Wheel

Steering Wheel Design







What?

- Developed a **custom steering wheel** for the Formula SAE vehicle, designed with a focus on **ergonomics** and **performance.**
- The project aimed to improve the driver's control by integrating force sensors and steering angle sensors for **real-time feedback** and optimal steering performance.
- Emphasized **precision manufacturing** using **waterjet cutting** and **3D printing** to create durable components, ensuring reliability and **sensor integration** into the steering system.

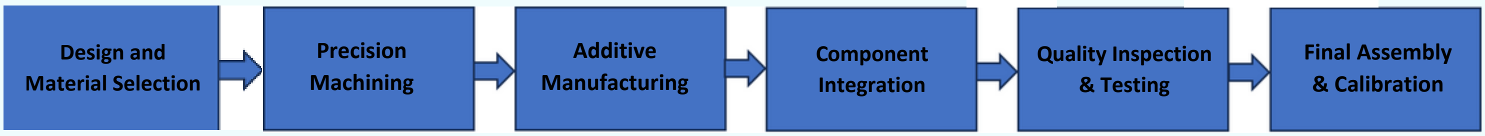
How?

- Designed the steering wheel using **SolidWorks** for precise **3D modeling**, optimizing sensor placement
- Utilized **waterjet cutting** for **high-precision manufacturing** ensuring strength and accuracy.
- Applied **3D printing** to create customized handles, enhancing driver comfort and control.
- Integrated **Raspberry Pi** and other sensors within the **waterjet-cut frame**, ensuring reliable performance and **easy assembly**.

Results

- Achieved a functional **steering wheel** system, integrating **motion tracking** and **sensor feedback** that interact with the vehicle's control unit to provide **real-time steering feedback** and enhanced **control accuracy**.
- Successfully implemented a **closed-loop control system** for steering, enabling the system to adjust based on vehicle dynamics and driver inputs.
- The final system showed improved **handling and precision**, conforming the integration of **sensors and control module** for the FSAE vehicle.

Steering Wheel Control System Manufacturing Process Flow



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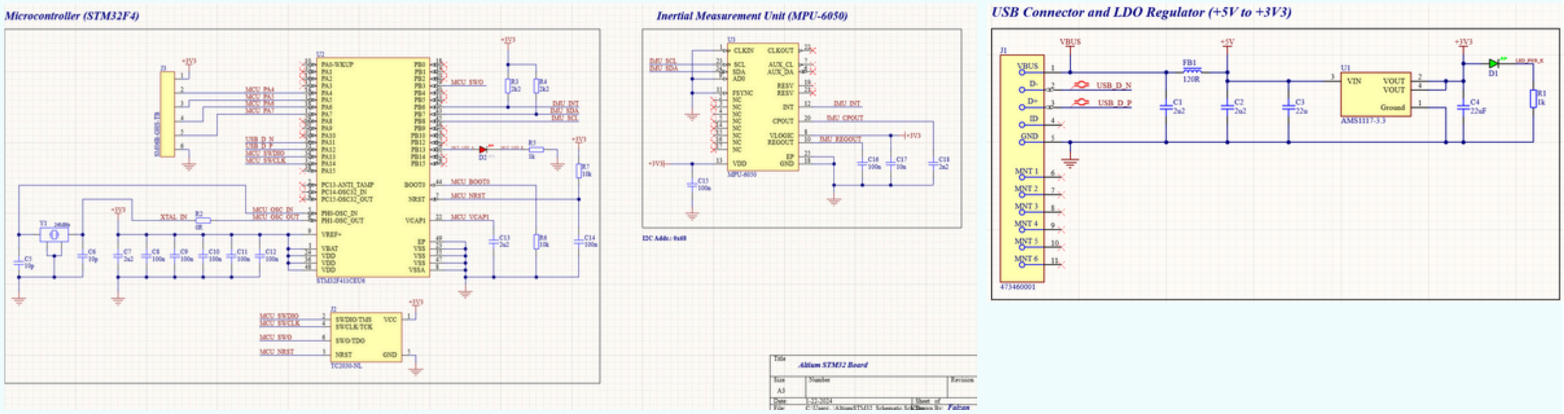
STM32 PCB Design Project (Altium Designer)

STM32 F411CEU6 Schematic



What?

- A custom-designed **PCB project** focused on creating a functional and compact board using the **STM32 F411CEU6 microcontroller**.
- The aim was to design a board capable of interfacing with various **sensors and modules** for numerous applications including **robotics** and **IoT** devices.
- The project serves as a foundation for developing advanced prototypes and **embedded systems** with a focus on performance and scalability.

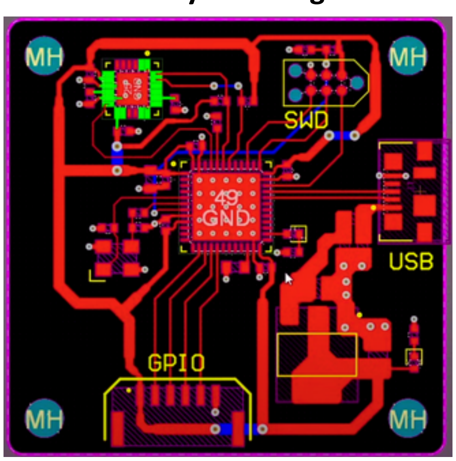
How?

- Utilized **Altium Designer** to **create** and develop and both **2D schematic** layouts and **3D PCB designs,** ensuring precision in electronic circuitry.
- Incorporated a range of components such as the MPU-6050 for **motion tracking**, **USB connectivity** for programming and **power**, and **GPIO** headers for peripheral expansion.
- Used best practices in PCB design, including **trace width** calculations, **impedance matching**, and **layer management** for signal integrity.

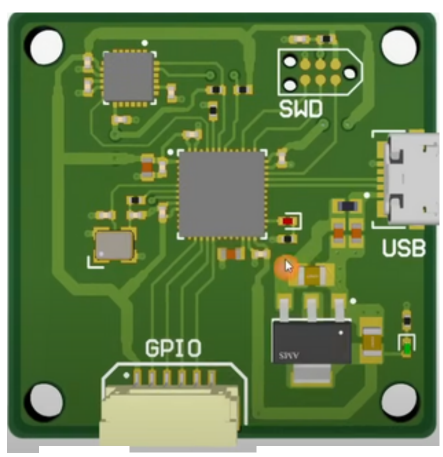
Results

- Achieved a **fully operational** STM32-based **PCB** with **integrated IMU** capabilities, capable of performing computational tasks and interfacing with external hardware.
- The project concluded with the **successful validation** of the PCB's design, resulting in a scalable prototype that demonstrated excellent potential for **commercialization** and real-world application deployment.

PCB Layout Design



Final 3D PCB Model



Component Selection → Schematic Design → PCB Layout Design → Design Finalization (DRC) → Preparation for Production (Gerber files)